

Module A1 Practice Exercises Workbook

AI assisted statistical work in the national information ecosystem

Candidate v1.0

Module OS-A01 | Asset OS-A01-A08 | Canonical lesson OS-A01-A01 | The National Information Ecosystem

This controlled asset must be used with the approved canonical lesson and canonical notebook v1.0.1.

Purpose and required submission

This workbook guides you through four linked practicals. You will establish a conventional baseline, use AI for bounded assistance, preserve the AI output, verify it independently, correct it, and decide whether the resulting workflow improves quality-adjusted performance. Completion requires professional judgement rather than acceptance of generated text.

- Submit this workbook and the notebook output folder.
- Preserve exact prompts, tool or model context, candidate outputs, corrections and elapsed time.
- Do not place protected or confidential information in an unapproved AI service.
- The statistician and subject-matter specialist retain non-delegable responsibility for definitions, methods, uncertainty, fitness and disposition.

Learner identification

Field	Response
Learner name	
Institution and unit	
Facilitator	
Date started	
Date submitted	
AI tool and version if used	
Notebook version	OS-A01-A01 v1.0.1

Assessment components

Component	Weight	Required evidence
Scope and statistical purpose	15%	Boundary, intended use, population,

Component	Weight	Required evidence
		product and exclusions
Actors assets and flows	20%	Stable, source-traceable registers
Quality risk and fitness	20%	Error mechanisms, user requirements and consequences
AI task allocation and prompting	15%	Bounded prompt, AI record and appropriate allocation
Verification and correction	20%	Independent checks, error log and accepted product
Judgement and communication	10%	Recommendation, limitations and oral defence

Professional rules and workflow

The seven stage evidence cycle

Stage	Learner action	Evidence retained
1 Authorize	Classify data, identify owners and approve tool conditions	Authorization record
2 Define	Set decision, boundary and acceptance criteria	Boundary statement
3 Baseline	Complete a representative task conventionally	Manual sample and time
4 Assist	Use a structured prompt or bounded AI tool	Exact prompt and raw output
5 Verify	Check claims, logic, citations and failure conditions	Verification record
6 Correct	Repair or reject material errors under named authority	Correction log
7 Compare	Assess accepted quality and total effort	Quality-adjusted conclusion

AI and human authority

AI may support	AI must not decide	Human authority
Structured extraction, comparisons, candidate code, test ideas and drafts	Legal authority, statistical meaning, representativeness, release fitness or final acceptance	Statistician, subject specialist, data owner and other formal reviewers decide within mandate

Authorization gate

Control	Learner record
Purpose	
Product owner	
Statistical reviewer	
Subject-matter reviewer	
Data classification	

Control	Learner record
Approved tool conditions	
Prohibited content	
Stopping rules	

Practical 1 Actor authority and evidence map

Estimated time: 75 minutes. Complete ACT-01 through ACT-04 conventionally. Use an approved AI tool or the controlled notebook candidate for the remaining records. Verify every authority claim against E01-E10.

Task 1 Define the boundary

Dimension	Included	Qualification or exclusion

Task 2 Conventional actor sample

ID	Actor	Role or roles	Authority or constraint	Evidence

Task 3 Structured prompt

Write the exact prompt. It must specify purpose, evidence boundary, output schema, prohibited inferences, uncertainty handling, acceptance criteria, human owner and stopping rule.

Prompt element	Your specification
Task	
Evidence boundary	
Output schema	
Uncertainty rule	

Prompt element	Your specification
Prohibited inference	
Acceptance criteria	
Stopping rule	
Human verification	

Practical 1 Candidate and verification record

ID	Candidate claim	Evidence cited	Check performed	Disposition

Correction log

Record	Material issue	Independent evidence	Correction or rejection	Human owner

Accepted actor register

ID	Actor	Verified roles	Authority or constraint	Evidence

ID	Actor	Verified roles	Authority or constraint	Evidence

Practical 1 reflection

Which AI claim was most plausible but wrong?

Response: _____

Which decision required statistical or subject-matter authority?

Response: _____

What changed between the candidate and accepted register?

Response: _____

Practical 2 Asset inventory and source validation

Estimated time: 90 minutes. Inventory the survey, district reports, satellite imagery, farmer registry and pilot results. Do not treat availability as proof of fitness. Record population, coverage, time, identifiers, transformation, quality and access conditions.

Asset inventory

ID	Asset and owner	Coverage and time	Identifiers and transformation	Quality and access	Evidence

Source validation plan

Asset	Risk domain	Verification question	Evidence or test	Human owner	Status

Practical 2 Total error and integration analysis

Source	Coverage risk	Measurement risk	Processing risk	Estimation or generalization risk	Required response

Integrated source assurance

Issue	Question	Proposed diagnostic	Acceptance authority
Linkage			
Duplication			
Temporal alignment			
Classification			
Uncertainty propagation			

AI contribution and specialist contribution

Work element	AI could do	AI should do under controls	Statistician or SME must do

Practical 3 Critical flow and failure analysis

Estimated time: 90 minutes. Build the structured flow register before drawing a diagram. Confirm the actual process with its owner. Record transformations, controls, delays, failure modes, fallbacks and escalation.

ID	Input and source	Transformation and output	Control	Failure and consequence	Fallback and owner

Risk prioritization

Flow	Likelihood 1 to 4	Consequence 1 to 4	Detectability 1 to 4	Priority rationale	Decision

Stopping and escalation rules

Observable condition	Immediate action	Escalation owner	Restart condition

Observable condition	Immediate action	Escalation owner	Restart condition

Practical 4 Fitness for use and AI pilot decision

Estimated time: 105 minutes. Convert user needs into explicit evidence requirements. Then assess candidate AI tasks against value, verifiability, consequences, readiness and human oversight. Include non-AI alternatives.

Dimension	Requirement	Acceptance evidence	Human owner	Status
Relevance				
Accuracy				
Reliability				
Timeliness				
Coherence				
Comparability				
Accessibility				
Interpretability				

AI opportunity comparison

Task	Value	Verifiable	Consequence if wrong	Readiness	Oversight	Decision

Prompt and interaction record

Field	Record
Task identifier	
Date and time	
Tool model and version	
Configuration or settings	
Approved source set	
Exact prompt	
Raw output location	
Reviewer disposition	
Corrections required	
Reproducibility limitation	

Quality adjusted efficiency comparison

Effort component	Conventional minutes	AI assisted minutes	Evidence or comment
Source preparation			
Prompt design			
Execution			
Verification			
Correction and rework			
Escalation			
Total accepted effort			
Accepted quality result			

Conclusion: AI-assisted workflow was ☐ faster and acceptable ☐ better quality ☐ neutral ☐ worse ☐ discontinued

Justification: _____

Professional recommendation

Decision element	Learner recommendation
Statistical problem and intended use	
Current workflow weakness	
AI task proposed or rejected	
Non-AI alternative	
Evidence supporting the decision	
Pilot boundary and acceptance criteria	
Verification and monitoring plan	
Named human authorities	
Stopping rules	
Residual limitations	

Oral defence preparation

- Identify one AI output you rejected and cite the evidence.
- Explain one correction that required subject-matter expertise.
- Defend one task that must remain non-AI.
- Explain whether time savings remained after verification and correction.
- State the conditions required before transferring this workflow to real institutional data.

Submission checklist

- Completed authorization and boundary
- Conventional baseline sample and time
- Exact prompts and raw candidate outputs
- Verification and correction evidence
- Accepted actor asset and flow records
- Fitness-for-use and opportunity decisions
- Quality-adjusted comparison
- Professional recommendation and residual limitations